



COMP30026

Models of Computation

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Lecture 3

Consequence and Satisfaction



Recap: Models

$\models$ is short for “is a model of” or “satisfies”.

$v \models F$ iff F is true under v .



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$$\{P \mapsto 0\} \models \neg P$$



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Examples:

$$\{P \mapsto 1, Q \mapsto 0\} \models P \vee Q$$

$$\{P \mapsto 0\} \models \neg P$$

Non-examples:

$$\{P \mapsto 1, Q \mapsto 0\} \not\models P \rightarrow Q$$

$$\{P \mapsto 1\} \not\models \neg P$$



Recap: Equivalence

“ $\equiv$ ” is short for “is logically equivalent to”.

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Non-examples:

$$\begin{aligned}P \rightarrow Q &\not\equiv R \rightarrow S \\ P \wedge Q &\not\equiv P \vee Q\end{aligned}$$



Semantic Consequence

Definition

G is a *semantic consequence* of F **if and only if** every model of F is a model of G .

For short, we write “ $F \models G$ ”.

“ $\models$ ” is pronounced “(semantically) entails”.



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For short, we write “ $F \models G$ ”.

“ $\models$ ” is pronounced “(semantically) entails”.

Note: $F \equiv G$ iff $F \models G$ and $G \models F$.



Poll

Of the following formulas, which allow us to conclude $P \rightarrow Q$?

1. P
2. $\neg P$
3. Q
4. $P \rightarrow (Q \wedge R)$
5. $(P \vee R) \rightarrow Q$
6. $\neg P \vee Q$
7. $\neg Q \rightarrow \neg P$
8. $P \rightarrow (Q \vee R)$
9. $(P \rightarrow Q) \vee R$



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Of the following formulas, which allow us to conclude $P \rightarrow Q$?

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Which of the formulas are consequences of $P \rightarrow Q$?



Tautology

Tautology: a logical formula which is always true.

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$(\neg P \wedge Q) \rightarrow (P \rightarrow R)$ is a tautology:

P	Q	R	$(\neg P \wedge Q) \rightarrow (P \rightarrow R)$						
1	1	1	0	1	0	1	1	1	1
1	1	0	0	1	0	1	1	0	0
1	0	1	0	1	0	0	1	1	1
1	0	0	0	1	0	0	1	0	0
0	1	1	1	0	1	1	1	0	1
0	1	0	1	0	1	1	1	0	0
0	0	1	1	0	0	0	1	0	1
0	0	0	1	0	0	0	1	0	0



Contradiction

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When P and Q are both true:

P	Q	$(P \wedge Q) \wedge (\neg Q \leftrightarrow (\neg P \vee Q))$									
1	1	1	1	1	0	0	1	0	0	1	1



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Negating a contradiction gives a tautology and vice versa.



Tautologies Are Valid

Consider: “If the program works, then the program works.”

It is **true** *regardless* of what “program” or “works” mean.



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It is **true** *regardless* of what “program” or “works” mean.

Valid: Always true.

Non-valid: False for some valuation.

$\models F$ is short for “ F is valid”.



Contradictions Are Unsatisfiable

Consider:

“the application is good and the application is not good.”

It is **false** *regardless* of what “the application” or “good” mean.



Contradictions Are Unsatisfiable

Consider:

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It is **false** *regardless* of what “the application” or “good” mean.

Unsatisfiable: Never true.

Satisfiable: True for some valuation.



Most Statements Are Contingent

Consider: “It is currently raining.”

It is true **if and only if** it is currently raining.



Most Statements Are Contingent

Consider: “It is currently raining.”

It is true **if and only if** it is currently raining.

Contingent: Sometimes true, sometimes false.



Consequence and Implication

Let F and G be formulas.

Theorem

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As an immediate corollary:

Corollary

$F \equiv G$ if and only if $\models F \leftrightarrow G$.



Poll

Classify the following formulas as valid, contingent, or unsatisfiable:

1. P
2. $P \leftrightarrow \neg P$
3. $P \rightarrow (\neg Q \vee P)$
4. $\neg Q \vee \neg(P \wedge \neg Q)$



Poll

Classify the following formulas as valid, contingent, or unsatisfiable:

- | | |
|--|-----------------|
| 1. P | ■ contingent |
| 2. $P \leftrightarrow \neg P$ | ■ unsatisfiable |
| 3. $P \rightarrow (\neg Q \vee P)$ | ■ valid |
| 4. $\neg Q \vee \neg(P \wedge \neg Q)$ | ■ valid |



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Replacing **all** uses of a propositional variable with a given formula.



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Consider " $P \rightarrow P$ ".



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Result: " $(Q \wedge R) \rightarrow (Q \wedge R)$ ".



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Example

Consider " $P \rightarrow P$ ".

Substitute P with " $(Q \wedge R)$ ".

Result: " $(Q \wedge R) \rightarrow (Q \wedge R)$ ".

Substitution preserves validity!



Poll

Does substitution preserve unsatisfiability?



Poll

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Yes!



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Yes!

Negation of contradiction is a tautology.



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No — a counterexample is easy:



Poll

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No — a counterexample is easy:

Take P (which is clearly satisfiable).

Then substitute P by $Q \wedge \neg Q$.



Substitution Preserves Logical Equivalence

Denote by $F[A := B]$ the result of substituting A with B in F .

Example: $(P \rightarrow P)[P := Q]$ is $(Q \rightarrow Q)$.

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Theorem

Let F, G, H be formulas and P be any propositional variable.

If $F \equiv G$, then $F[P := H] \equiv G[P := H]$.



Interchange of Equivalents

If $F \equiv G$, then F can be freely replaced with G .



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Theorem

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If $F \equiv G$, then $H \equiv H'$.

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Result is equivalent: all semantic properties preserved.

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Theorem

Let H' be the result of replacing an instance of F with G in H .

If $F \equiv G$, then $H \equiv H'$.

Result is equivalent: all semantic properties preserved.

Rewrite formulas algebraically!

Some Equivalences

Absorption:

$$P \wedge P \equiv P$$
$$P \vee P \equiv P$$

Commutativity:

$$P \wedge Q \equiv Q \wedge P$$
$$P \vee Q \equiv Q \vee P$$

Associativity:

$$P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$$
$$P \vee (Q \vee R) \equiv (P \vee Q) \vee R$$

Distributivity:

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$
$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

More Equivalences

Double negation: $P \equiv \neg\neg P$

De Morgan: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
 $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Implication: $P \rightarrow Q \equiv \neg P \vee Q$

Contraposition: $\neg P \rightarrow \neg Q \equiv Q \rightarrow P$
 $P \rightarrow \neg Q \equiv Q \rightarrow \neg P$
 $\neg P \rightarrow Q \equiv \neg Q \rightarrow P$

Biimplication: $P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q)$



Last Equivalences

$\perp$ is equivalent to any unsatisfiable formula

$\top$ is equivalent to any valid formula

Duality: $\neg \top \equiv \perp$
 $\neg \perp \equiv \top$

Negation from absurdity: $P \rightarrow \perp \equiv \neg P$

Identity: $P \vee \perp \equiv P$
 $P \wedge \top \equiv P$

Dominance: $P \wedge \perp \equiv \perp$
 $P \vee \top \equiv \top$

Contradiction: $P \wedge \neg P \equiv \perp$

Excluded middle: $P \vee \neg P \equiv \top$



Poll

Which of these claims hold?

1. $P \rightarrow Q \equiv (Q \leftrightarrow (P \vee Q))$
2. $(P \rightarrow Q) \wedge (P \rightarrow R) \equiv P \rightarrow (Q \wedge R)$
3. $(P \rightarrow R) \wedge (Q \rightarrow R) \models (P \wedge Q) \rightarrow R$



Poll

Which of these claims hold?

1. $P \rightarrow Q \equiv (Q \leftrightarrow (P \vee Q))$ Yes
2. $(P \rightarrow Q) \wedge (P \rightarrow R) \equiv P \rightarrow (Q \wedge R)$ Yes
3. $(P \rightarrow R) \wedge (Q \rightarrow R) \models (P \wedge Q) \rightarrow R$ Yes, but not $\equiv$